Requirements

Loan Approval System

A **digital lending platform** for a mid-to-large financial institution modernizing its retail loan approval process. This is a realistic, high-stakes domain that touches every dimension of enterprise AI/ML architecture.

Business Context

The bank processes thousands of loan applications daily. Legacy systems are slow, manual-intensive, and unable to scale with demand or meet digital-first customer expectations.

The Opportunity

An AI-powered decisioning layer can dramatically reduce approval cycle times, improve credit risk accuracy, and enable straight-through processing for low-risk applications.

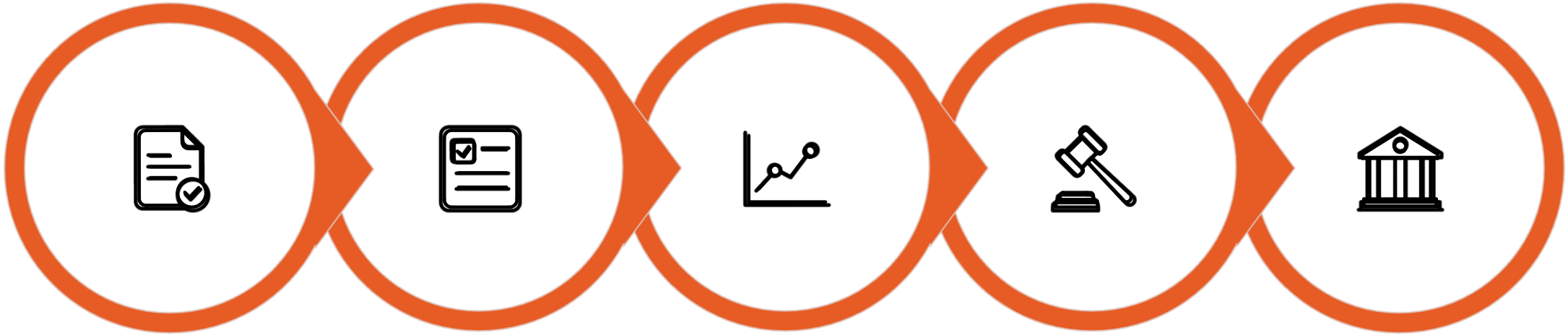
Key Drivers

Speed: Customers expect near-instant decisions. **Accuracy:** Better ML models reduce default rates. **Compliance:** All decisions must be explainable and auditable under financial regulations.

SECTION 3 — CASE STUDY

Business Process Overview

The loan approval process follows a clear sequential flow — each stage is a potential integration point for AI/ML intelligence. Understanding this flow is the foundation for all architecture decisions.



Application **Validation** **Underwriting** **Decision** **Funding**

Notice that stages 1 and 2 are largely rule-based, stage 3 is a clear AI/ML opportunity, and stage 4 involves a hybrid of model output and business rules. Architecture must serve all of these — not just the AI parts.

SECTION 3 — CASE STUDY

Problem Statement

Before designing the solution, it's critical to clearly define what's broken. The current state has four compounding problems:

High Manual Effort

Credit analysts manually review the majority of applications. This is slow, expensive, and introduces human inconsistency. Analyst capacity — not demand — is the bottleneck.

Slow Approval Cycles

Average time-to-decision is measured in days, not seconds. In a digital-first market, this is a significant competitive disadvantage and a source of customer churn.

Risk of Defaults

Rule-based credit scoring fails to capture behavioral signals and complex risk patterns. Default rates are higher than peer benchmarks, directly impacting profitability.

Lack of Intelligent Decisioning

The system cannot learn from outcomes. There's no feedback loop connecting approval decisions to repayment behavior — meaning the model never improves.